



Portolan chart
From the 13th century Mediterranean sailors used a distinctive kind of map, the portolan. Coasts and ports were well charted and the maps were criss-crossed with a network of lines projecting from compasses.

funds to develop powerful navies to defend their trade routes and fight their competitors—chiefly one another. There was in this period only relatively minor conflict between Christians and Muslims at sea, but intense armed rivalry between Christian states, much of it fought out around the coasts of Italy and Sicily between galleys from Catalonia and Provence as well as various Italian ports.

VENETIAN EMPIRE

Of all medieval Mediterranean states, Venice was the quintessential naval power. The shipbuilding and repair facilities of the Venetian Arsenal, founded around 1200, grew into the biggest industrial enterprise in Europe—it was said they could build and fit out a galley in a single day. Venetian galleys were rowed "alla sensile"—with oarsmen grouped in threes, each with a single oar—and proved successful, highly maneuverable warships. The Venetians were generally reckoned the first to adopt such innovations as the compass, the stern rudder, and the lateen sail. Their oarsmen—volunteers or conscripts—were honored seamen rather than galley slaves, some earning promotion through long careers to captain. By the 15th century Venice had created a Mediterranean empire based on sea power, as Athens had done in ancient times. But by the end of that century profound changes were under way, both in the naval balance of power with the rise of the Ottoman Empire, and in technology with the introduction of cannon into naval warfare.

Provisioning the Crusades
The larger items of cargo are being loaded aboard a cog, while the galley in front of it is filled with armed men. Among the banners of the various crusading states involved, those of England and France are most prominent, along with the crossed keys of the papacy.

